

AirGrid - Smart Wireless Infrastructure Planning Platform

AirGrid is a web-based wireless infrastructure intelligence platform designed for smart campuses, buildings, and urban wireless planning. The system helps users place wireless access points, detect interference, identify dead zones, and optimize signal coverage through an interactive planner.

1. Core Features

Feature	Purpose
Interactive Planner Canvas	Grid/floor layout where users place wireless nodes
Add Devices	Add WiFi AP, Router, IoT Node, BLE Beacon
Drag & Drop Nodes	Move devices and update coverage live
Coverage Circles	Show each node's wireless range
Signal Heatmap	Visualize signal intensity
Interference Detection	Detect overlap between APs
Dead Zone Detection	Highlight uncovered areas
Channel Conflict Detection	Warn if nearby APs use same channels
Metrics Dashboard	Coverage %, dead zones, interference score
Smart Suggestions	Suggest AP movement/additions/channel changes
Save/Load Projects	Store layouts in MongoDB
Generate Reports	Create professional network reports

2. Advanced Features

Feature	Benefit
Wall/Obstacle Mode	Walls reduce signal strength realistically
Auto Optimize Button	Suggest best AP placement automatically
Frequency Selection	2.4GHz / 5GHz / BLE support
Node Configuration Panel	Configure range, power, channel
Mini-map	Overview of the entire layout
PDF Export	Generate professional reports
AI Explanations	Gemini-generated recommendations

3. Recommended Tech Stack

Layer	Technologies
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Frontend	React + Vite + Tailwind + React Konva + Framer Motion
Backend	Node.js + Express.js
Database	MongoDB Atlas + Mongoose
Charts & Analytics	Recharts
Icons/UI	Lucide React
AI Integration	Gemini API
PDF Generation	html2canvas + jsPDF

4. Navbar Structure

- Dashboard** → Project overview and quick actions
- Planner** → Interactive wireless infrastructure canvas
- Analytics** → Coverage, interference, and signal metrics
- Optimizer** → Smart AP suggestions and optimization engine
- Reports** → PDF export and deployment summaries
- Projects** → Saved layouts and previous analyses

5. Recommended UI Layout

- Top Navbar:** AirGrid branding + navigation tabs
- Left Sidebar:** Device tools, toggles, controls, actions
- Center Canvas:** Interactive grid with APs, heatmaps, signal overlays
- Right Panel:** Contextual analytics and AP details
- Bottom Metrics Strip:** Live network statistics and warnings

6. MVP Build Order

1. Planner Canvas
2. Add + Drag AP Nodes
3. Coverage Visualization
4. Dead Zone Detection
5. Interference Detection
6. Metrics Dashboard
7. Smart Suggestions
8. Save/Load Projects
9. Reports & PDF Export
10. Advanced Optimization Features